FUNCTION AND REGULATION OF THE GI TRACT

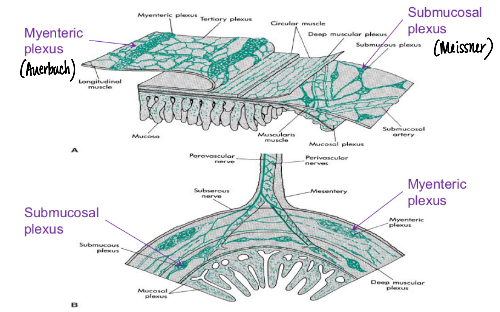
Integrated Control of GI functions

Ingestion
↓
Secretions (7-10L daily) ⇒ Digestion + Absorption ⇐ Motility
↓
Elimination

GIT structure (microscopic)

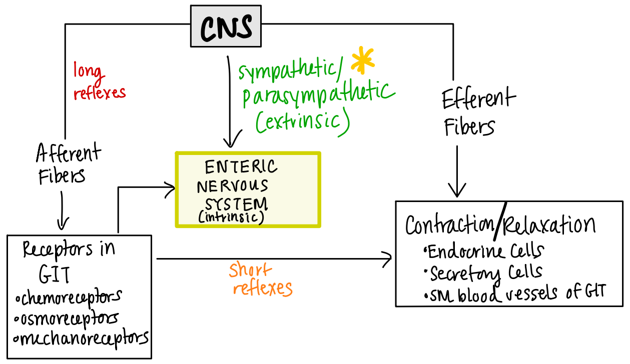
Epithelial Cells: specialized for various secretions and absorption of nutrients.
Muscularis Mucosae: contractions change the surface areas for secretion/absorption.
Circular Muscle: contractions decrease the diameter of the lumen
Longitudinal Muscle: contraction shortens the segment of the GIT.
Submucosal Plexus (Meissner) + **Myenteric Plexus**: Integrate the motility, secretory, and endocrine functions of the GIT.

The Enteric Nervous System



ENS- arranged anatomically into nerve networks in the myenteric + submucosal plexuses.
↳ 100 million neurons (sensory, inter, motor neurons)
↳ concerned soley with regulating GI function.
ENS can function autonomously without extrinsic connections.
Myenteric Plexus: involved with control of motility + innervates the longitudinal/circular smooth muscle layers. (Muscularis Externa)

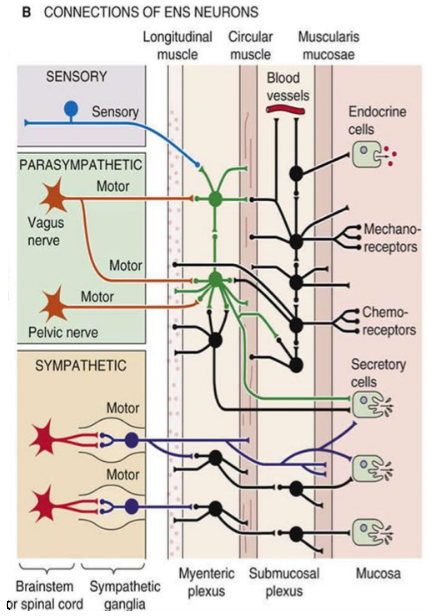
Submucosal Plexus: innervates the glandular epithelium, intestinal endocrine cells + submucosal blood vessels to control intestinal secretion.
-recieves sensory info. from chemo + mechanoreceptors.



Extrinsic Innervation of the GIT

Parasympathetic
- excitatory
- Vagus (upper GIT) and pelvic nerves (lower GIT)
- preganglionics synapse on plexi
- postganglionics synapse on smooth muscle, secretory, and endocrine cells

Sympathetic
- inhibitory
- Thoracoculumbar fibers (T8-L2)
- preganglionics (cholinergic) synapse in the prevertebral ganglia
- post-ganglionic synapse on plexi
- plexi → relay info to target cells



Basically...

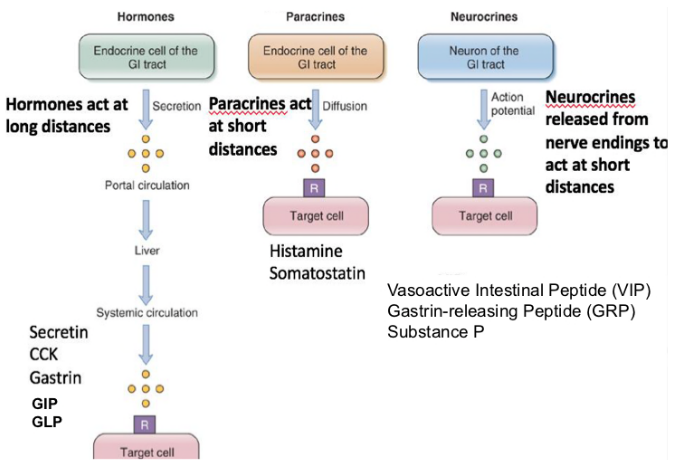
- Short reflexes integrated in ENS
- Long reflexes integrated in CNS
- Connections between CNS/ENS via ANS

Neural, hormonal, and paracrine regulation

Neurocrine:
- peptides released by nerves in the gut
- short-range effects after release
- Ach, Substance P (SM contraction)
- Norepinephrine, vasoactive intestinal peptide (relaxation)

Endocrine:
- Hormones synthesized by cells in the gastric + intestinal epithelium and released into the blood stream
- systemic long-range action
- Gastrin, secretin, CCK, incretins (GIP + GLP)

Paracrine:
- Substances that act locally within the same tissue
- short-range effects
- Histamine, somatostatin



Hormonal Regulators

Gastrin: secreted from the gastric antrum after a meal
- Stimuli for release:
  - ↳ small peptides + amino acids (especially phenylalanine + tryptophan) in stomach
  - ↳ stomach distension
  - ↳ vagal stimulation, mediated by gastrin-releasing peptide (GRP)
  - ↳ gastrin secretion is inhibited by increasing H^+ (acidity) in lumen of stomach.
  - ↳ gastrin also inhibited by somatostatin.
- effects
  - ↳ increases H^+ secretion by gastric parietal cells
  - ↳ stimulates growth of gastric mucosa

Secretin: 27 amino acids, homologous to glucagon
- Effects
  - reduces acidity of small intestines
  - stimulates pancreatic HCO_3^- secretion (neutralizes H^+ in the intestinal lumen)
  - Stimulates HCO_3^- + H_2O secretion by the liver and increases bile secretion
  - Inhibits H^+ secretion by gastric parietal cells
- Stimuli for release (by duodenal S-cells)
  - H^+ in lumen of the duodenum
  - Fatty acids in the lumen of the duodenum

Cholecystokinin (CCK): structurally homologous to gastrin
- Effects:
  - gallbladder contraction and relaxation of the sphincter of oddi for bile secretion.
  - Stimulates pancreatic enzyme secretion
  - potentiates secretin-induced stimulation of pancreatic HCO_3^- secretion
  - Inhibits gastric emptying → meals containing fat stimulate the secretion of CCK, which slows gastric emptying to allow more time for intestinal digestion + absorption.
- Stimuli for release: (from I-cells of the duodenal + jejunal mucosa)
  - small peptides and amino acids
  - fatty acids + monoglycerides

Incretins: GIP + GLP

GIP: gastric inhibitory peptide (Jejunum K-cell)
GLP: glucagon-like peptide (GLP-1) (Colon L-cell)
both enhance insulin secretion in response to oral carbohydrates

- The total amount of insulin secreted is greater when glucose is administered orally than when administered intravenously, hence digestion is at least partly driven by anticipatory response.
- GIP also inhibits H^+ secretion by the gastric parietal cells, limiting acidification by the stomach.

Paracrine Regulatory Mediators

Somatostatin:
- secreted throughout the GIT in response to luminal H^+
- secretion is inhibited by vagal stimulation
- inhibits the release of all GI hormones
- inhibits gastric H^+ secretion

Histamine:
- secreted by mast cells of the gastric mucosa
- increases gastric H^+ secretion directly and by potentiating the effects of gastrin + vagal stimulation.

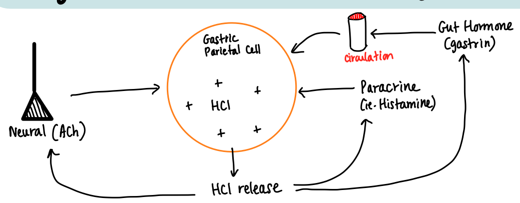
Neurocrine Regulatory Peptides

Vasoactive Intestinal Peptide (VIP) homologous + functionally similar to secretin.
- released by neurons in the GIT mucosa + smooth muscle
- produces relaxation of GI smooth muscle, including lower esophageal sphincter
- stimulates pancreatic HCO_3^- secretion
- Inhibits gastric H^+ secretion

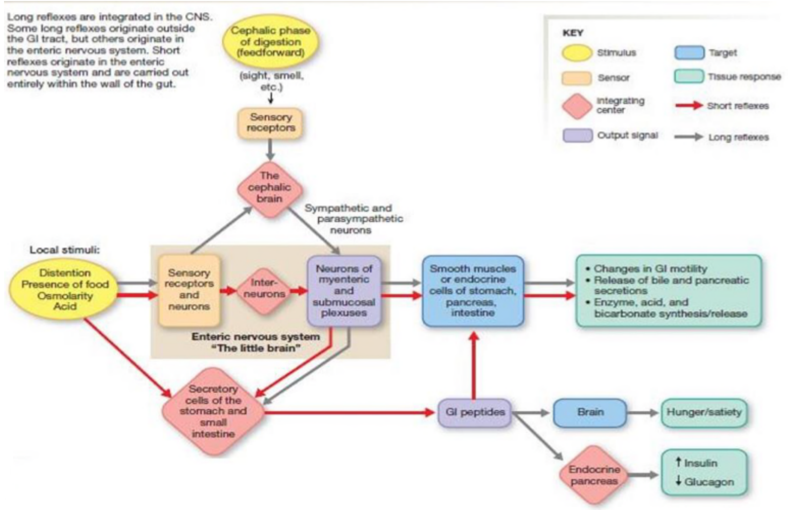
Gastrin-releasing peptide (GRP, also bombesin)
- released from vagus nerve that innervates the G-cells.
- stimulates gastrin-release from G-cells.

Enkephalins (met-enkephalins + leu-enkephalins)
- secreted from nerves in the mucosa and smooth muscle in the GIT.
- Stimulates contraction of GI smooth muscle, particularly the:
  - lower esophageal, pyloric, and ileocecal sphincters.
- Inhibit intestinal secretion of fluid + electrolytes, which forms the basis for the usefulness of opiates in the tx of diarrhea.

Integration + Overlap of Regulators



End organ responses involve integration and overlap of signaling modes - neural, hormone, and paracrine.



Stages of Meal Processing

① Cephalic: anticipation, sight, smell, + taste of food

- ↑ salivary secretion
- ↑ gastric, pancreatic and intestinal secretions
- ↑ motility

Mediated by long reflexes:

- Afferents: sight, smell, food in mouth
- Coordinating centre: Medulla Oblongata
- Effectors: Vagus nerve, ENS

② Gastric: when food is in stomach

- Stimulus: stomach distension and presence of peptides + amino acids in stomach.
- coordinating center: Local endocrine cells + ENS.
- Effectors: Hormones, paracrines, and neurotransmitters
- Actions: ↑ secretions + motility
- Results:
  - digestion of proteins in stomach by pepsin
  - formation of chyme
  - controlled by entry of chyme into the SI.

③ Intestinal: when food is in intestine

- Trigger: Entry of chyme into the SI
- Actions: A series of reflexes that aid the entry of chyme into SI to promote digestion and motility.

Central appetite regulation

Leptin "satiety hormone" → secreted by fat cells.
- stimulates neurons that decrease appetite and inhibit those that increase appetite.
- insulin + GLP-1 also inhibit appetite

Ghrelin "hunger hormone" – secreted by gastric cells. ↑s appetite.

Both: act on feeding centers of the hypothalamus which integrates the various signals into an overall hunger/satiety feeling.

ENDOCRINE

Peptide	(+) secretion	(-) secretion	Site of Secretion	Target Cells	Action
Gastrin	AA in stomach, stomach distension, Gastrin releasing peptide, Ach (Vagus)	Acid pH<1.5, Somatostatin	**G** cells of antrum, duodenum, jejunum	Entero-chromaffin like cell(ECL) Parietal cells	Stimulates gastric acid ↑ gastric mucosal growth
CCK	FA, AA, distension	Somatostatin	**I** cells of duodenum jeunum and ileum	Secretion in circulation	*Motility of gall bladder *reduced delivery of chyme from stomach *↑pancreatic secretions *↑satiety
Secretin	Acid in SI	Somatostatin	S cells of duodenum, jejunum and ileum	Pancreas Stomach	↓gastric acid ↑HCO3 secretion by pancreas Neutralizes acid delivered to duodenum Optimum pH for intestinal enzymes
Glucose dependent insulin-tropic polypeptide GIP *GLP-1	FA, AA, CHO	Neuropeptides	K cells of duodenum and jejunum Upper GI	β Cells of pancreas	Stimulates insulin release Inhibits gastric acid secretion

NEUROCRINE

Peptide	(+) secretion	(-) secretion	Site of Secretion	Target Cells	Action
Vasoactive Intestinal peptide	Released by action potential	Somatostatin	Neurons in the mucosa and smooth muscle of the GIT	Smooth Muscles including Lower esophageal sphincter	• Relaxation of the GI smooth muscle • Pancreatic HCO_3^- secretion • Inhibits gastric H^+ secretion
Gastrin Releasing Peptide	Vagus nerve	Acid in the stomach	Post ganglionic fibers of Vagus nerve that innervate the G cells	G cells	*Stimulates gastrin secretion

Paracrine

Peptides	(+) secretion	(-) secretion	Site of Secretion	Target Cells	Action
Somatostatin	H+ in the lumen	Vagal stimulation	**D cells of stomach and**	Potent Inhibitor	➢(↓) pancreatic /gastric secretion ➢(↓)motility ➢(↓)gall bladder contraction ➢(↓)nutrient absorption ➢vasoconstrictor
Histamine	Food in the stomach	H+ in stomach	Entero-chromaffin like cell ECL, mast cells	Specific receptors	*Stimulates gastric acid secretion Enhances vascular vasodilation